

EEG-based ADHD Classification During a Visual-Attention Task

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Introduction and Research Question

Attention-deficit/hyperactivity disorder (ADHD) is associated with difficulties in attention and cognitive control. In this project, we investigate whether EEG recorded during a visual-attention task contains information that can distinguish children with ADHD from healthy controls at the participant level.

The dataset contains EEG recordings from 121 children, including 61 children with ADHD and 60 controls. EEG was recorded from 19 channels. During the experiment, children viewed images containing cartoon characters and were asked to count them. The next image appeared immediately after the child's response. Consequently, the total recording duration depended on task performance and response speed. This creates an important methodological issue: recording length may itself contain behavioral information and must not accidentally become a shortcut for the EEG classifier.

Research question: Can participant-level ADHD status be predicted from EEG recorded during a visual-attention task and how much discriminative information is provided by time-domain, frequency-domain and coarse temporal-position features?

We formulate this as a supervised binary classification problem. The aim is to evaluate whether EEG representations from this task contain reproducible group-related information under participant-level cross-validation, rather than to develop a clinical diagnostic system.

Dataset Description

The dataset [1] contains EEG recordings from 121 children, including 61 participants with ADHD and 60 healthy controls. EEG was recorded from 19 scalp channels at a sampling rate of 128 Hz and band-pass filtered between 0.5 and 40 Hz. According to information provided by the researchers upon request, no additional artifact-removal or rejection procedures, such as the removal of eye-movement or muscle artifacts, were applied. Likewise, no specific procedures for removing bad channels or noisy segments, re-referencing, segmentation/epoching, or normalization were performed. No recordings or participants were excluded during preprocessing. The researchers further indicated, upon request, that no additional information is available regarding the exact timing of the image presentations. The recordings were primarily intended to investigate participants' engagement in the arithmetic task rather than responses to individual image presentations.

The recordings include a participant identifier and a diagnostic label (ADHD or Control).

The ADHD participants were diagnosed according to DSM-IV criteria. The dataset description also reports that participants with ADHD had taken Ritalin for up to six months, which may represent a potential confound.

Recording length varied substantially between participants. ADHD participants contributed a mean of 19,788 EEG rows (median 18,389), compared with 15,989 rows (median 15,629) for controls. Because the next image was presented after each response, this difference may partly reflect differences in task performance and response speed. Recording length was therefore not used as a predictive feature.

Methods

Participant-safe evaluation

All data splits were performed at the participant level. Windows from the same child are never placed in different folds. This prevents participant leakage, where a model could see one part of a child's recording during training and another part during evaluation.

The final performance estimate uses stratified 5-fold participant-level cross-validation. In each outer fold, approximately four-fifths of the participants are used for training and one-fifth for evaluation. Within each outer fold, the outer-training participants were further split into training and validation sets. The validation set was used for early stopping and selection of the number of training epochs. After epoch selection, the model was reinitialized and retrained on all outer-training participants before predicting the untouched outer fold.

All preprocessing parameters, including feature standardization, are estimated using training participants only and then applied to validation or test participants.

Windowing and control of recording-length imbalance

Each participant's chronological EEG recording is divided into windows of 512 samples with a stride of 256 samples. Under the reported 128 Hz sampling frequency, this corresponds to approximately four-second windows with 50% overlap. If more than 80 windows are available for a participant, 80 starting positions are selected evenly from the beginning to the end of the recording. This limits the influence of participants with longer recordings while retaining windows from across the recording. The complete run yielded 7,625 windows across the 121 participants.

Because shorter recordings can still contain fewer windows, the MLP loss uses participant-equal weights: the total contribution of each participant's windows is approximately equal during training. Recording length itself is excluded from the EEG feature set. These safeguards reduce the influence of unequal recording duration, but they do not fully remove the underlying behavioral confound because response speed and EEG activity were observed during the same task.

Feature engineering

For each channel and window, six time-domain statistics are calculated: mean, standard deviation, minimum, maximum, median and root mean square (RMS). This results in 114 time-domain features across the 19 channels.

To represent spectral information, the discrete Fourier transform is used to estimate log power in five frequency ranges: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and low gamma (30–40 Hz). These produce 95 frequency-domain features.

Finally, each window is assigned a normalized relative position between 0 and 1, indicating its position within the participant's recording. This variable retains coarse information about task position but does not model the full sequential dependence between windows.

Models and implementation

A logistic-regression classifier using the same time- and frequency-domain features as the main MLP serves as a linear baseline.

The required neural network is a feedforward multilayer perceptron implemented from scratch with NumPy. The forward pass, ReLU hidden activations, sigmoid output, binary cross-entropy loss, backpropagation, Adam optimization and L2 regularization are implemented directly rather than using a pre-built neural-network estimator. The network is intentionally kept small because the number of independent participants is limited. The initial baseline configuration used two hidden layers with 64 and 32 units, a learning rate of 0.0005 and L2 regularization of 0.001.

The same MLP architecture is evaluated with four feature sets:

- time-domain features only;
- frequency-domain features only;
- combined time- and frequency-domain features;
- combined time-domain, frequency-domain and relative-position features (additional ablation configuration).

The final diagnosis is made at participant level. The MLP first predicts an ADHD probability for every window and probabilities from all windows belonging to one participant are averaged. Participant-level accuracy, balanced accuracy, precision, recall, F1-score and ROC-AUC are reported. Window-level observations are not treated as independent clinical cases.

Hyperparameters and learning behavior

The prespecified MLP used two hidden layers with 64 and 32 units, a learning rate of 0.0005, L2 regularization of 0.001, mini-batches of 64 windows and a maximum of 80 training epochs. The number of training epochs was selected separately within each outer fold using the inner participant-level validation set. Participant-level validation F1 was used as the selection criterion, and early stopping prevented unnecessary training once validation performance no longer improved. The outer test fold was used only for final evaluation.

The MLP used mini-batches of 64 windows and a maximum of 80 training epochs. Hyperparameters were then tuned for the main MLP configuration using only the inner training and validation participants within each outer fold. Eight predefined configurations varying hidden-layer size, learning rate and L2 regularization were evaluated. The configuration with the highest participant-level validation F1 was selected. The selected configuration was then retrained on all outer-training participants before evaluation on the held-out outer fold.

Within each outer fold, the number of training epochs was selected using the inner participant-level validation subset. Participant-level validation F1 was used as the selection criterion, and early stopping prevented unnecessary training once validation performance no longer improved. Training loss, validation loss, validation participant-level F1 and validation ROC-AUC were recorded for every epoch and used to generate the learning curves. The outer test fold was used only for final evaluation.

All values are from the complete participant-level five-fold cross-validation run with `QUICK_MODE = False`.

As a confound diagnostic, a participant-level logistic-regression model using recording length alone achieved an accuracy of 0.653 and ROC-AUC of 0.655. Recording length was therefore not included as a predictive feature.

The logistic-regression baseline using the time- and frequency-domain features achieved pooled participant-level accuracy 0.711, F1-score 0.711 and ROC-AUC 0.796.

Among the MLP configurations, the time + frequency model achieved the strongest participant-level performance, with pooled accuracy 0.769, balanced accuracy 0.768, precision 0.754, recall 0.803, F1-score 0.778 and ROC-AUC 0.832. Across the five folds, mean F1 was 0.780 ± 0.102 and mean ROC-AUC was 0.826 ± 0.065 (see [Appendix Table A1](#)).

For the selected time + frequency MLP, early stopping selected 5, 9, 2, 9 and 7 epochs across folds 1–5, respectively (range 2–9; see [Appendix Figure A1](#)).

Feature ablation

The time-only MLP achieved F1 0.748 and ROC-AUC 0.793, while the frequency-only MLP achieved F1 0.645 and ROC-AUC 0.659. Combining time- and frequency-domain features increased performance to F1 0.778 and ROC-AUC 0.832, while adding normalized task position reduced performance to F1 0.738 and ROC-AUC 0.810.

Adding frequency-domain features to the time-domain representation increased F1 by 0.030 and ROC-AUC by 0.039. Using frequency features alone resulted in a larger decrease relative to the time + frequency model ($\Delta F1 = -0.133$, $\Delta \text{ROC-AUC} = -0.173$). Adding relative task position reduced F1 by 0.040 and ROC-AUC by 0.022. Overall, spectral features provided additional discriminative information when combined with time-domain features, whereas coarse task position did not. Fold-to-fold variability remained substantial (see [Appendix Figure A2](#) and [Appendix Table A1](#)).

The pooled ROC curve and participant-level confusion matrix for the selected time + frequency MLP are shown in [Appendix Figure A3](#).

Discussion

Participant-level five-fold cross-validation showed that combined time- and frequency-domain features provided the strongest MLP representation (accuracy 0.769, F1 0.778, ROC-AUC 0.832). Time-domain features alone were already informative, whereas frequency-domain features alone were substantially weaker. The improvement from time-only to time + frequency suggests that spectral information contributed complementary discriminative information when combined with amplitude- and variability-based summaries.

Adding normalized relative window position did not improve performance (F1 0.738, ROC-AUC 0.810). Thus, the coarse indication of whether a window occurred earlier or later in the task did not provide additional predictive value beyond the EEG-derived time- and frequency-domain features in the present analysis.

The recording-length diagnostic deserves separate consideration. Recording length alone achieved accuracy 0.653 and ROC-AUC 0.655, showing that task duration contained some group-related information. Because the experiment advanced after each child's response, this likely reflects behavioral variation in response speed as well as acquisition duration. Recording length was therefore excluded from the EEG classifier; the analysis also capped and evenly distributed windows and used participant-equal training weights. These safeguards reduce, but cannot fully eliminate, the possibility that task performance and EEG activity are intertwined.

Several limitations remain. The dataset contains only 121 participants, so cross-validation reduces dependence on a single split but does not remove sampling uncertainty. Windows from the same participant are correlated, which is why participant-level metrics are treated as primary. The ADHD group had taken Ritalin for up to six months, medication may therefore confound group differences and the recordings come from a single visual-attention task and dataset. Frequency-domain results depend on the reported 128 Hz sampling rate and the selected band definitions. Cross-validation estimates internal generalization within this dataset; it does not establish external or clinical validity.

The exploratory channel analysis also should not be interpreted causally. P7 showed the largest average absolute effect size (mean $|d|$ = 0.552), followed by T7, Pz, P3 and Fz, but these group-level differences do not establish that any individual channel is diagnostic or specific to ADHD.

Overall, the results support the conclusion that EEG representations recorded during this visual-attention task contain group-related information that generalizes to held-out participants within this dataset. The model should be interpreted as an exploratory classifier, not as a clinical diagnostic tool.

A possible extension of the present approach would be to apply an EEG-specific convolutional architecture such as EEGNet [\[2\]](#). In the current analysis, the MLP operates on manually engineered time- and frequency-domain summary features. EEGNet instead learns

representations directly from multichannel EEG using temporal convolutions to capture frequency-related information and depthwise convolutions to learn frequency-specific spatial filters across electrodes. Lawhern et al. [2] designed EEGNet as a compact architecture for EEG classification and showed competitive performance across several EEG paradigms, including settings with limited training data and cross-subject evaluation. This is relevant to the present dataset, where the number of independent participants is relatively small. EEGNet could potentially capture temporal and spatial relationships that are lost when each window is reduced to summary statistics. However, the EEGNet study investigated BCI paradigms rather than ADHD classification, so improved performance on the present task cannot be assumed and would need to be evaluated using the same participant-level cross-validation procedure.

Appendix

1. Figures

1.1 Figure A1: Learning curves

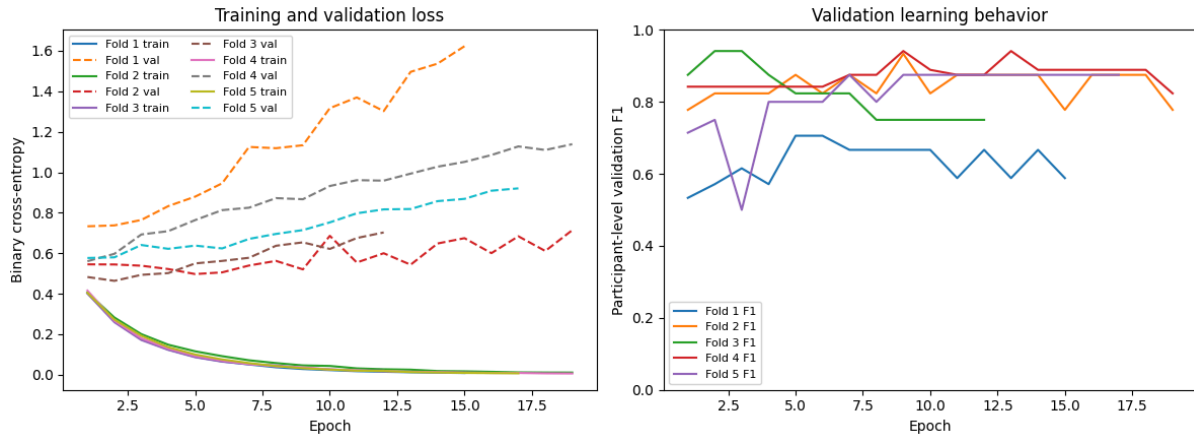


Figure 1. Training and validation behavior of the selected time + frequency MLP across the five participant-level folds.

1.2 Figure A2: Model comparison

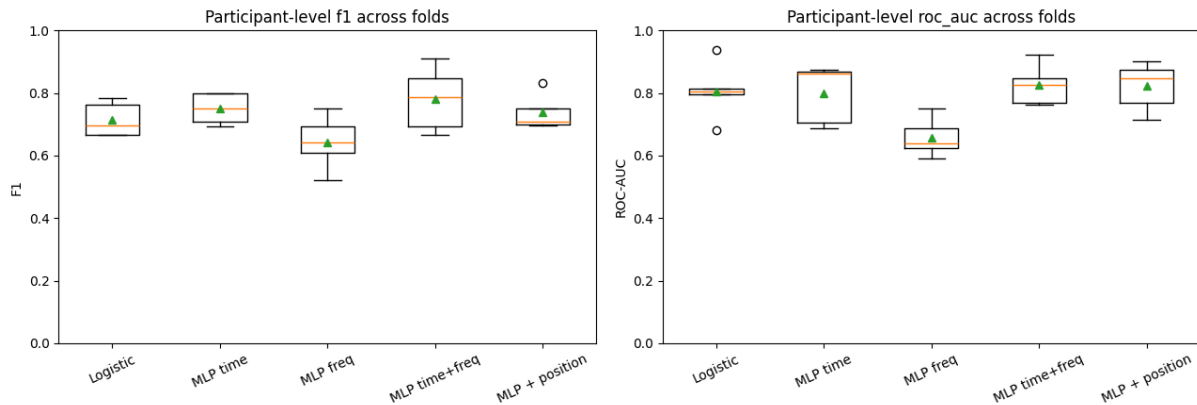


Figure 2. Participant-level F1 and ROC-AUC across the five cross-validation folds for the logistic baseline and four MLP feature configurations. Green triangles indicate fold means; orange lines indicate medians.

1.3 Figure A3: ROC curve + confusion matrix

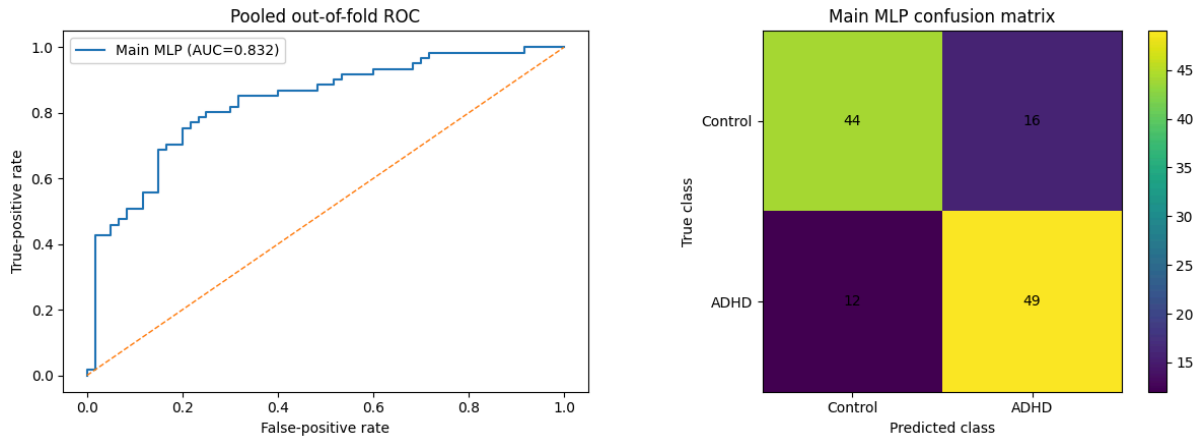


Figure 3. Participant-level out-of-fold performance of the selected MLP using combined time- and frequency-domain EEG features. Left: pooled ROC curve across the five cross-validation folds (AUC = 0.832). Right: participant-level confusion matrix at a probability threshold of 0.5. The model correctly classified 44 of 60 controls and 49 of 61 ADHD participants; 16 controls were classified as ADHD and 12 ADHD participants as controls.

1.4 Figure A4: Channel effect sizes

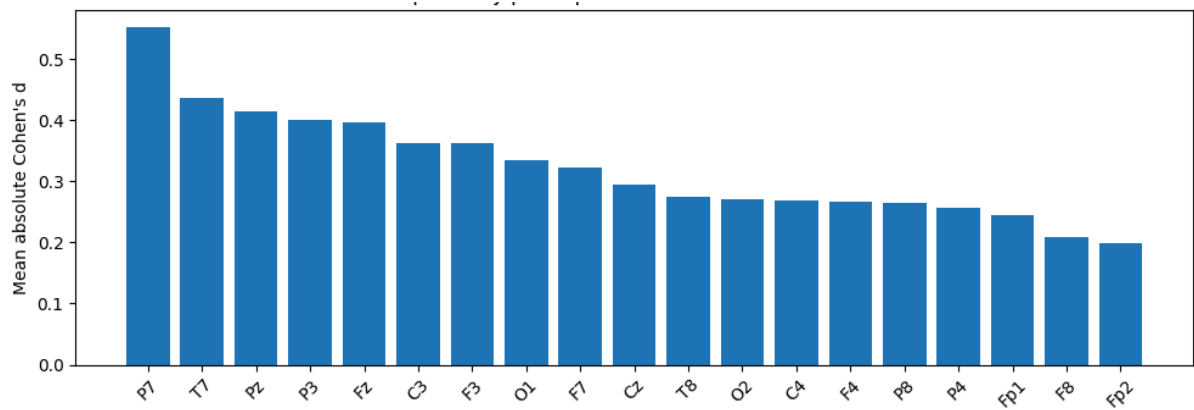


Figure 4. Exploratory participant-level channel differences, summarized as mean absolute Cohen's d across features belonging to each EEG channel.

2. Tables

2.1 Table A1: Cross-validation performance

	model	f1	roc_auc	delta_f1_vs_time	delta_auc_vs_time	delta_f1_vs_time_frequency	delta_auc_vs_time_frequency
1	mlp_time	0.748	0.793	0.000	0.000	-0.030	-0.039
2	mlp_frequency	0.645	0.659	-0.103	-0.134	-0.133	-0.173
3	mlp_time_frequency	0.778	0.832	0.030	0.039	0.000	0.000
4	mlp_time_frequency_position	0.738	0.810	-0.010	0.017	-0.040	-0.022

Table 1. Participant-level cross-validation performance of the baseline and MLP feature configurations.

References

- [1] D. Nizo, "EEG Dataset for ADHD," *Kaggle*. Available: <https://www.kaggle.com/datasets/danizo/eeg-dataset-for-adhd>
- [2] Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018). *EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces*. *Journal of Neural Engineering*, 15(5), 056013. <https://doi.org/10.1088/1741-2552/aace8c>